Academic Performance Prediction of Students Using Explainable Machine Learning Ensembles Techniques

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Abstract-Prediction of students' academic performance has become an important part of the contemporary educational system due to its capacity to detect students who need further help when studying. Even though the application of conventional methods of machine learning to predict academic performance may yield good results, their inability to be explained hinders implementation in practice. This paper proposes to use the idea of ensemble learning in machine learning to predict students' academic performance. Specifically, two classifiers will be used, which will be combined into a single predictive model based on the soft voting algorithm. To provide explainability of the predictive model, the SHAP method is applied. The practical evaluation of the proposed solution is implemented based on real-life data on students' academic performance, and the advantages of the proposed predictive model are disclosed. Lunch type, test preparation, and parental education are among important factors affecting academic performance of students.

Keywords-Students' Performance Prediction, Ensemble Learning, Explainable AI, SHAP, XG Boost, Random Forest

I. INTRODUCTION

Prediction of the academic success of students in modern education systems is important because it enables one to detect the students that are academically at risk and take preventive actions. Contemporary advances in technology make it possible to collect and analyze large amounts of data on the subject matter [1], [16], [17]. EDM uses machine learning methods to extract meaningful information from the available data and improve academic performance of students.

Machine learning methods use such methods as decision trees, logistic regression, and naïve Bayes classifiers for predicting student academic performance. Even though these methods generate quite accurate predictions, they are not capable of identifying relationships between the variables [2]. On the other hand, Random Forest and boosting algorithms have proven to be superior [3], [20].

Nevertheless, although the performance of most ensembles is satisfactory, most models function as black boxes, which makes the whole process difficult to comprehend. To address this challenge

One of the first approaches in XAI includes SHAP (SHapley Additive exPlanations), which provides explanations about single prediction scenarios [4], [18]. In this connection, the major concern of this work will be the development of an XAI model based on the ensemble technique for prediction of the students' academic success.

II. LITERATURE REVIEW

The early literature on predicting students' outcomes mainly focused on statistical analysis and simple machine learning algorithms. The research stressed the importance of educational data mining for improving the learning experience and making intelligent decisions within the organization [1], [16].

Prediction of students' outcomes was also performed using classification approaches, but such an approach lacked some limitations when handling large data sets [2].

With the advent of modern machine learning algorithms, there is significant improvement in prediction accuracy. For example, employing ensembling algorithms, such as the random forest algorithm, yielded high accuracy by utilizing multiple decision trees to prevent overfitting [3]. In addition, modern machine learning algorithms, such as gradient boosting through the application of XG Boost, have been applied to ensure high accuracy due to efficient model computation and regularization [5].

Comparison between traditional and machine learning approaches to predict student outcomes has been studied. It has been shown that the machine learning models yield higher accuracy compared to traditional approaches due to their robustness [11], [20]. Besides, the machine learning approaches yield impressive results if relevant features of student behavior and academics are considered [12], [17].

However, the problem of interpretability is just as relevant as that of sufficiency of accuracies. Speaking about the goal of applying the concepts of Explainable Artificial Intelligence, their application improves the process of interpretation of the ensemble. First of all, it should be said that the main principle of the SHAP algorithm function is based on game theory concepts. Hence, the value of contribution of every feature in relation to the score can be assessed [4], [18]. Secondly, there have been a number of cases where application of SHAP helped to identify important variables related to education, as well as such features as parents' education, time spent on studying, and socio-economic status of individuals [6].

There are also some approaches that are used to increase interpretability. One of the approaches is an approach of interpretation of local prediction outputs. Trade-off between interpretability and accuracy in machine learning is

considered one of the most urgent issues [13], [14], [19]. Despite the number of successful cases of work in this field, not much focus has been directed at the design of a unified approach that would combine interpretability and ensemble learning methods. This is why this research will attempt to integrate the three aforementioned methodologies into one model.

III. SUGGESTED APPROACH TO DATA ANALYSIS

The proposed approach was already used for analyzing the publicly available datasets that could be accessed by any researcher. The dataset included students' information about their academic achievements and socio-demographic characteristics [1],[16].

However, there were some initial actions that needed to be taken for receiving good results from machine learning modeling. The issue was that the dataset had both categorical and numerical features, and therefore some steps had to be made [1]. For example, categorical features like gender, parent education, lunch, and test preparation could be transformed into numerical data [7].

In addition, all the numerical features had to be normalized [19]. Moreover, it was required to divide the dataset into two parts, where one dataset should consist of 70% and the other one – of 30%.

A. Explanation of Dataset and Data Preprocessing

As was mentioned above, this machine learning technique is implemented to the above-mentioned open source dataset which consists of data concerning academic performance as well as socioeconomic background of students. Educational data mining researches are usually conducted on the basis of open source datasets [1].

However, the preprocessing of datasets prior to using any machine learning algorithm is essential. In this case, transformation is needed because it contains both numerical and categorical variables. The first ones are represented by such features as gender, parents' education level, lunch type and test preparation course [7].

Also, all numerical values are normalized to make equal contribution to the improvement of machine learning model performance. Besides, ratio between training and testing data sets should be set to 70% and 30% correspondingly. Education level, lunch type, and test preparation course become numeric [7].

B. Our contributions

• The key objective of our study is to propose a machine learning model that is interpretable and can predict students' academic achievement by using the notion of ensemble learning [20].

• We used the approach of soft voting with Random Forest classifier and XG Boost classifier for our predictions [3],[5],[20].

• Performance of our model will be evaluated using performance parameters such as accuracy, precision, recall, and f-measure [10],[17].

• The interpretation of our predictions will be done with the help of SHAP [4],[18].

D. Overall Workflow of the System

The diagram below shows how the flow of process works with regard to this algorithm. First of all, the initial process starts with collecting of the data set showing the performance of the students. After that, the process moves to data preprocessing where the data will be encoded if it consists of categorical data while normalized for numeric data. In this process, it should be highlighted that the data will be split into two data sets - testing and training data sets.

Secondly, the process moves to training of the machine learning algorithms such as random forest and XGBoost.

Overall System workflow

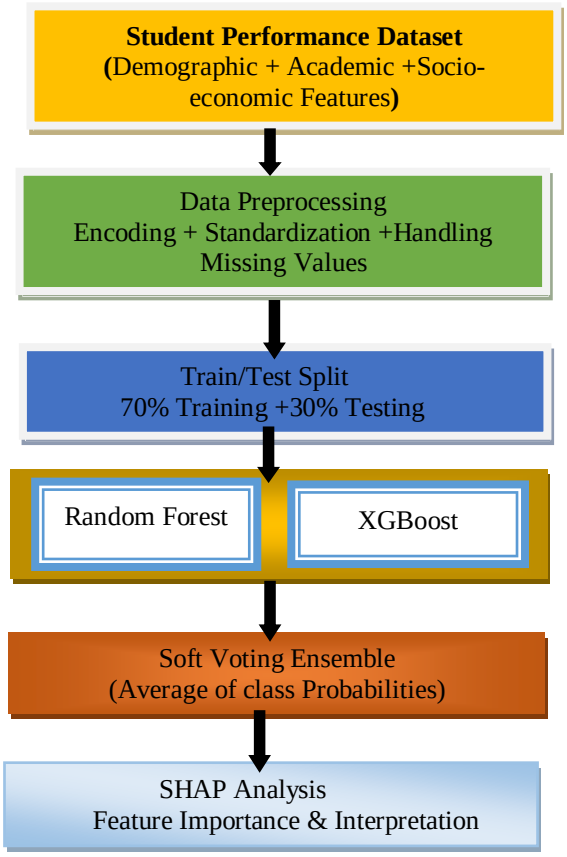

Fig. 1: Overall system workflow for student academic performance prediction using explainable ensemble learning.

E. Ensemble Learning Algorithm and Explanation Strategy

The above figure highlights the inner components of the ensemble algorithm. The input features like gender, parental level of education, and lunch programs are used for the models Random Forest and XG Boost. Each model's output results in probability scores for the class labels (Pass or Fail). The outputs of both the models are combined using an approach known as "soft voting" [8], [20].

Afterward, SHAP is applied to determine the feature importance values [4], [18].

Simplified Ensemble Explainability Framework

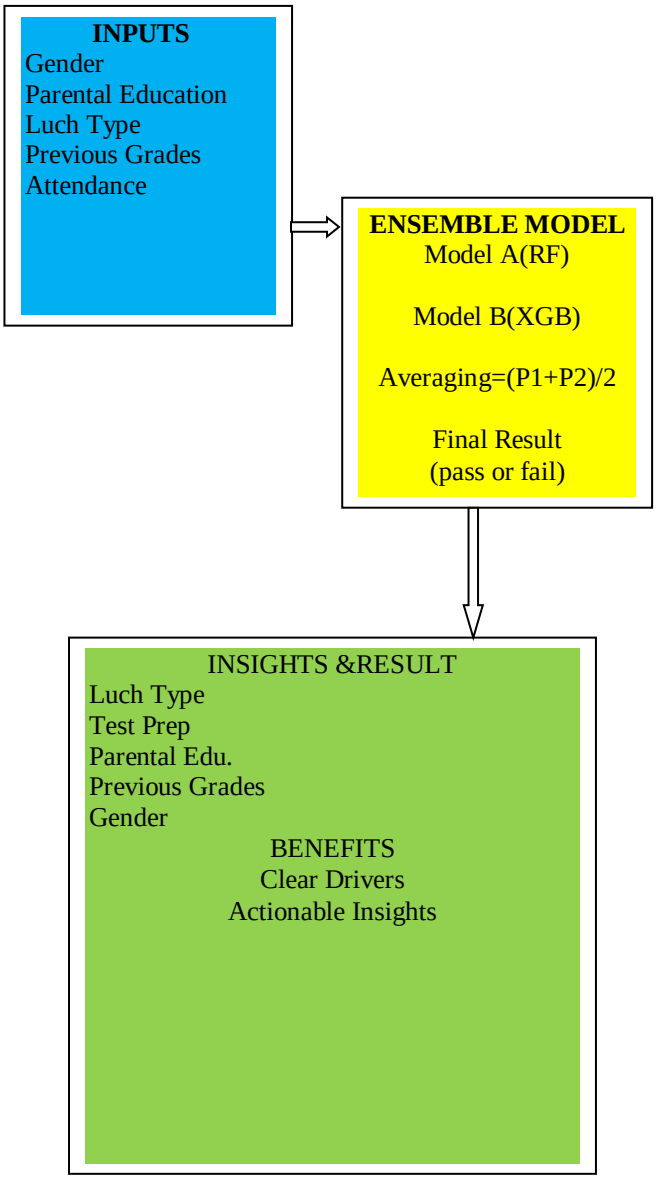

Fig. 2: Ensemble learning and explainability model based on Random Forest, XGBoost, soft voting, and SHAP explainability model.

1) Random Forest Classifier

Random forest constructs numerous decision trees using bootstrap aggregation and feature selection techniques to reduce variance and increase precision [3].

TABLE I: PERFORMANCE METRICS FOR RANDOM FOREST CLASSIFIER

Metric	Class	Class Completed	Overall Accuracy
Precision	0.73	0.58	0.68
Recall	0.82	0.45	-
F1-Score	0.77	0.51	-
Support	129		200(Total)

2) XG Boost Classifier

The XG Boost Classifier uses the gradient boosting approach to create decision trees step by step with an optimal objective function. The XG boost classifier has produced better outcomes for problems related to classification based on structured data [5].

TABLE II: RANKING OF FEATURES BASED ON FEATURE IMPORTANCE USING XGBOOST

Rank	Feature	Importance Score
1	reading score	0.31
2	Writing score	0.27
3	test preparation course	0.13
4	Parental level of education	0.11
5	Luch	0.090
6	Race	0.06
7	gender	0.03

3)Soft Voting Ensembe

In soft voting, the probabilities of classes are taken in the mean based on the output of the two classifiers (Random Forest, XGBoost) [8].

D. Equations

The equation of the provided dataset is illustrated below:

$X = \{x_1, x_2, \dots, x_n\}$

Here, 'x' represents feature vector and 'y' represents the class labels.

Final prediction for the ensemble model is made using following equation:

$P(y) = (1/N) \sum P_i(y)$ (1)

Here, N is the number of classifiers, and $P_i(y)$ is the output probability from i-th classifier.

E. Performance Measurement

Accuracy, Precision, Recall, F1 Score, Confusion Matrix [10], [17].

F. Explanation Using SHAP

For more transparency in the explanation of the results of ensembles, the SHAP (SHapley Additive exPlanations) method is applied. It is based on game theory and relies on feature importance metrics [4], [18], [19].

In this way, not only local, but also global understanding of the prediction procedure is achieved.

IV. RESULTS AND DISCUSSION

In this section, the experimental results obtained from the proposed ensemble-based model for the prediction of student academic achievements are analyzed. The model performance will be assessed using traditional evaluation criteria, whereas SHAP-based explainability analysis will be used for interpretation purposes and identification of the most significant variables.

The confusion matrices shown in Figs. 3 and 4 demonstrate the classification results of the proposed ensemble-based model for the prediction of student academic achievements. From the confusion matrices, one can notice that the model provides outstanding performance, classifying many students into the Pass class and several into the Fail class.
Figure 3 shows the classification when the true predicted result for the Pass class included 163 records, while for the Fail case, the true prediction included only 19 records, but some errors remained (8 false positives and 10 false negatives). As shown in Figure 4, the increase in model performance is evident since the number of true predicted results increased to 168 records for the Pass class and to 7 for the Fail class despite some inaccuracies, particularly for the second case.

Thus, the results of the model's performance indicate that its performance level is quite good because of high accuracy alongside successful predictions of both Pass and Fail classes.

Furthermore, Fig. 5 illustrates the SHAP Summary Plot. This plot facilitates the comprehension of how different variables contribute to the prediction. The variables are listed from top to bottom based on their importance levels. A single point in the figure is each student's record, where the x-axis denotes the SHAP values. Positive SHAP values in the x-axis denote higher chances for students being passed, while negative SHAP values imply the possibility of failing.
From the figure, it can be deduced that the variables 'lunch' (standard) and 'test preparation course' play a critical role in making predictions. Students with standard lunches and those who completed their test preparation courses get positive SHAP values. The contributions of the variable 'parental level of education' and races/ethnicities are moderate, whereas the variable 'gender' contributes minimally.
The SHAP interaction values of gender and race/ethnicity in relation to the model output are shown in Figure 6 below.
As seen in Figure 6, each observation refers to one student, while the SHAP interaction value along the X-axis denotes the sum total effect of two variables on the model prediction, considering the marginal effect. The smaller the SHAP interaction value, the lesser the interaction effect on the prediction of the model.
As seen from Figure 6 above, there is little to no existence of data points beyond the origin, indicating that SHAP interaction value does not significantly influence the model output.
Below is a graph representing the importance values received while predicting student academic performance through the XG Boost algorithm.
As one can see from the graph above, a standard lunch plays the biggest role among all other predictors. It provides an idea of the importance of the social and economic benefits received by students through lunch and the effect on their performances. Not attending preparatory classes for tests is the second most important predictor, thus emphasizing the role of preparation.
Moreover, parent's education level, such as some college, bachelor degree, high school, and master's degree, is essential when it comes to predictive model building. Speaking about the role of demographic predictors like racial or ethnic classification or gender of students, it seems like they do not play a prominent role in the model.
Therefore, it is clear from the above graph that XG Boost predictive model focuses on those factors which have a direct connection to student's performance.

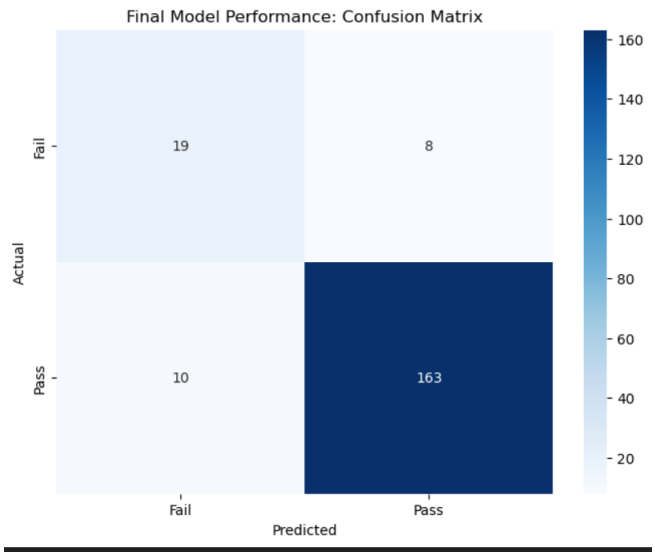

Fig. 3: Confusion matrix of the proposed ensemble model for student academic performance prediction.

TABLE III: Classification performance of the proposed ensemble model

Performance Metric	Class	Class Completed	Combined (Weighted)
Precision	*0.92*	*0.89*	*0.91*
Recall	*0.94*	*0.86*	*0.91*
F1-Score	*0.93*	*0.87*	*0.91*
Overall Accuracy			*91.00%*

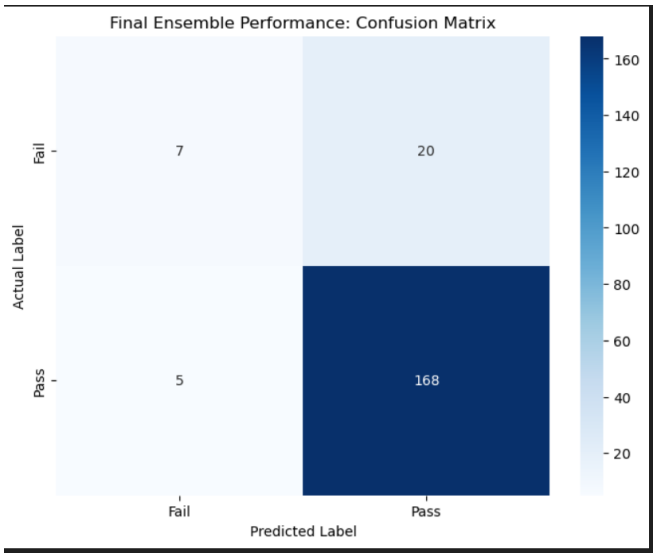

Fig. 4: Confusion Matrix of Classification Outcome from the Final Ensemble Model to Predict Whether Students Are Passed or Failed.

TABLE IV: PERFORMANCE EVALUATION OF THE PROPOSED ENSEMBLE MODEL

Metric	*Precision*	*Recall*	*F1-Score*
None	*0.86*	*0.94*	*0.93*
Completed	*0.89*	*0.86*	*0.87*
Overall Accuracy	-	-	*0.91*
Macro Average	*0.90*	*0.90*	*0.90*

TABLE V: IMPORTANT CHARACTERISTICS THAT INFLUENCE THE PERFORMANCE OF THE STUDENT ACCORDING TO XGBOOST

Model Name	Mean AbsoluteError	Root Mean Squared Error	Root Mean Squared Error
Linear Regression	0.000	0.000	1.000
Ridge Regression	0.055	0.071	0.999
Elastic Net	0.754	1.052	0.975

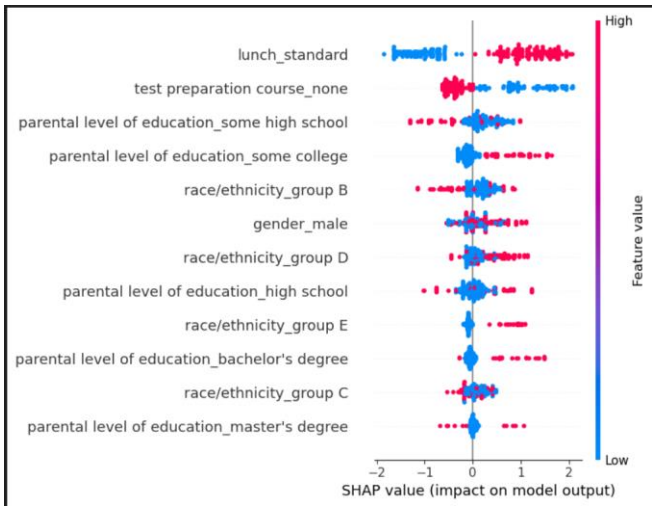

Fig. 5: Plot of SHAP Values Demonstrating Feature Importance and Impact on Prediction Outcomes.

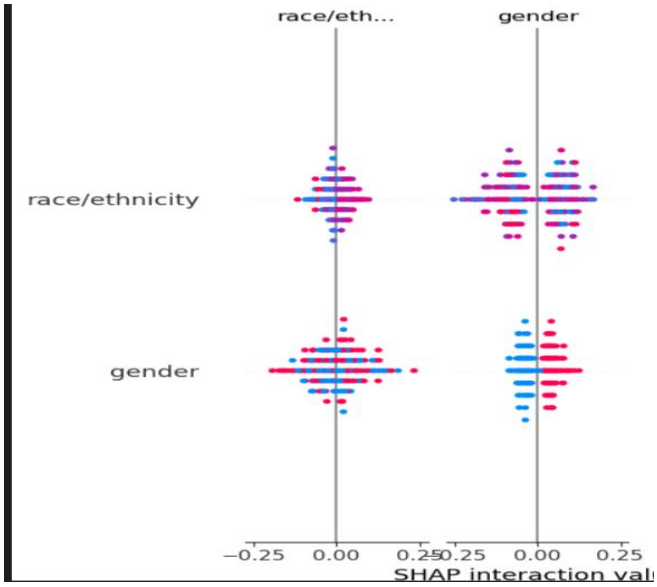

Fig. 6: Interaction Plot of SHAP Values Showing Gender Race/Ethnicity Interaction Effect on Prediction Outcomes.

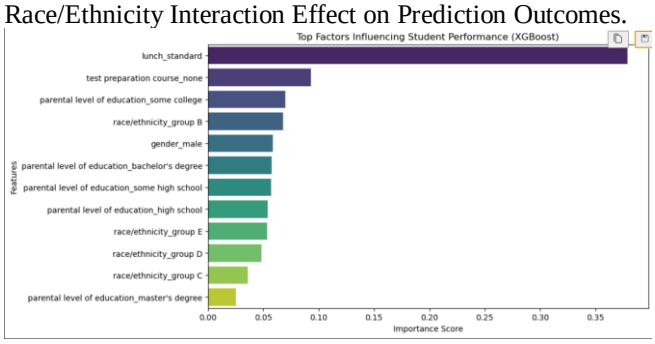

Fig. 7: Feature Importance Plot Generated Using the XG Boost Model Revealing Key Variables Influencing Students' Academic Achievement.

V. CONCLUSION

To conclude, the discussed paper introduced the concept of an explainable ensemble-based machine learning model to predict the academic success of students. This model consisted of the ensemble of Random Forest and XG Boost classifiers where the two mentioned models have been combined via soft voting as part of an ensemble technique. As for the problem of non-transparency in case of using machine learning algorithms, the suggested solution implied applying the method of explainability techniques based on SHAP. From this perspective, it is possible to state that the conducted empirical research based on the real-world data set on the topic under consideration provided decent results both from the viewpoint of classification and explainability. It was discovered that the level of effectiveness of classification together with its efficiency in terms of separation between different categories of students (whether to pass or fail) was fairly high. In addition, what concerns explainability, it became clear that the main predictors of whether a student will succeed include lunch, test preparation, and parents' education.

VI. FUTURE SCOPE

Despite the fact that the suggested framework has the potential to yield promising results, there are many ways in which the framework can be improved to enhance the performance of the framework further. For instance, when it comes to research in the future, using longitudinal data and behavioral data could be helpful in understanding the learning behavior over time and making accurate predictions. In addition to this, it should be noted that one of the potential improvements in the suggested framework involves the use of regression models rather than classification models for GPA prediction. The model must be unbiased.

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